

25PY101 (S2): Engineering Physics

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Mock pre M1T1 (Set I)

Instructions:

1. Answer all questions.
 2. Total marks – 10 M.
 3. Charge of electron is 1.602×10^{-19} C.
 4. Mass of electron is 9.1×10^{-31} kg.
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1 Metals and Semiconductors – Drift motion

- (a) What is the independent electron postulate of classical free electron theory (CFET)?
[1 M]
- (b) Derive the relation between the drift velocity v_d and electric field E . What is the constant of proportionality called in v_d vs E ? [3 M]
- (c) List two demerits and two demerits of CFET. [1 M]

2 Quantum mechanics – Particle picture of radiation

- (a) A blue lamp emits light of mean wavelength of $4500 \text{ \AA}$. The lamp is rated 150 W and 8% of the energy appears as light. How many photons are emitted per second by the lamp? [3 M]
- (b) Work function of sodium is 2.3 eV . Obtain the maximum wavelength which will cause emission of photoelectrons from the material. What will be the maximum kinetic energy of the photo-electrons emitted from the surface exposed to the radiation of $2000 \text{ \AA}$. [2 M]

End of Test